Iterative Progressive Retrieval Attention: Bounded Associative Closure over Latent Gist Memory

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Abstract

One-shot semantic retrieval finds memory nearest to the current query. It can miss a second piece of evidence that is weakly related to the query but strongly related to the first piece. We study iterative Progressive Retrieval Attention (PRA), which treats cached routing gists as an implicit graph: the query retrieves a bounded frontier, retrieved gists query the same layer-local index, and only the final closure is mapped to full native key/value (K/V) state. The transformer is not rerun between retrieval rounds.

A parameter-free implementation enforces depth, branch, beam, and unique-parent budgets and emits a retrieval graph before K/V materialization. Controlled two- and three-hop graphs validate chunk closure exactly, but inherited natural-data results are mixed. We therefore test two specific failure modes without retraining. First, the learned router is asymmetric: $z_q = W_q h_q$ and $z_m = W_m h_m$. Replacing the incorrect hop-2 frontier $W_m h_A$ by $W_q h_A$ recovers 0.075 QASPER any-evidence recall relative to historical closure at 10%, yet remains 0.013 below one-shot; HotpotQA chain completion is unchanged. Second, we encode 256-token parents once, derive eight 32-token local gists from each contextual state, traverse local nodes, and deduplicate parents before materializing native K/V. This local method solves every designed bridge control, where parent closure always fails. Finally, semantic narrowing followed by tokenwise layer-27 pre-RoPE query–key (QK) matching solves all 320 native-bridge controls. On natural data at a matched 20% parent budget, however, the primary native-QK condition reaches HotpotQA chain completion 0.125 versus 0.125 local semantic and 0.138 one-shot; QASPER reaches 0.900 versus 0.925 and 0.938. A Top-4 control reaches 0.175 on HotpotQA but falls to 0.838 on QASPER and does not dominate across budgets. Native propagation is therefore expressively sufficient on designed bridges but not a dependable pretrained associative geometry on these natural samples. One-shot remains the default, and any learned associative extension now has a clearer justification and baseline.

1 Introduction

Suppose a question is directly related to memory chunk A. Chunk A, in turn, identifies an entity or relation needed to find chunk B. A one-shot router computes similarities from the question to all chunks. It can retrieve A while missing B because the question does not yet contain the intermediate relation. Increasing top-k may find B, but it spends the same extra budget on unrelated chunks. The problem is not transformer reasoning after evidence arrives. It is evidence discovery before full memory becomes active.

PRA already separates compact semantic routing state from full layer-native K/V payloads [8, 7, 9]. This separation suggests a narrow extension. After retrieving A, use A's compact gist as another query. Continue for a bounded number of rounds, preserve the discovery graph, and materialize exact K/V only once. The resulting path $q \to A \to B$ is explicit. It does not require another language-model pass, an intermediate text answer, or serialization through an external tool API.

This paper asks whether pretrained hidden-state gists already possess enough transitive semantic geometry for that extension to help. The question has a strict baseline: iterative closure must beat one-shot Top-B at the same final unique-chunk budget B. Otherwise, retrieval depth adds cost without adding a useful primitive.

We keep the pretrained backbone and Paper 2 routing/transport stack fixed and replace one-shot Top-B discovery with parameter-free iterative closure. Projection correction, local-gist propagation, and native-QK propagation likewise alter discovery state only; they do not train the backbone or a memory-consumption adapter.

The contributions are sixfold. First, we formulate iterative PRA as query-conditioned bounded associative closure over an implicit latent graph. Second, we implement Level-1 gist-to-gist and Level-2 aggregate-frontier traversal with deterministic budgets, cycle handling, root anchoring, transparent path scores, and a versioned graph artifact. Third, we evaluate five semantic-router seeds on held-out HotpotQA and QASPER features, together with exact two- and three-hop synthetic controls. Fourth, we identify and correct an asymmetric projection-interface error without retraining. Fifth, we separate contextual encoding, local propagation, and parent K/V materialization scales, showing exact local-bridge recovery but no dependable natural-data chain gain. Sixth, we test semantically narrowed, tokenwise pre-RoPE native QK propagation under the same final parent budget, separate semantic and native search costs, and identify a controlled success but a natural-data boundary.

2 From Nearest Neighbors to Associative Closure

2.1 Why one-shot retrieval can stop too early

Let $G = \{g_1, \ldots, g_N\}$ be compact routing gists at one transformer layer and let q be the current routing query. One-shot PRA returns

$$N_B(q) = \mathrm{TopB}_{g \in G} \; s(q, g), \tag{1}$$

where s is cosine similarity in the learned semantic space. This operation is appropriate when every useful chunk is independently similar to q. It is incomplete when

$$s(q, g_A) \gg 0, \qquad s(q, g_B) \approx 0, \qquad s(g_A, g_B) \gg 0. \tag{2}$$

Here A is an entry point and B is relationally accessible only after A is known.

We interpret each chunk as a node in an implicit graph. High gist similarity supplies candidate directed edges, generated lazily when a node becomes part of the frontier. No $N \times N$ adjacency matrix is built. Iterative PRA therefore computes a bounded semantic neighborhood, not a proof and not complete reasoning.

2.2 The complete data path

Figure 1 separates traversal from payload activation. This boundary is the main systems invariant: compact gists may participate in several search rounds, while full K/V cannot enter attention until the closure stops.

ONE SHOT $q \to \{A, X, Y\} \to$ full native K/V

ITERATIVE ASSOCIATIVE CLOSURE

$$q \to A \to B \to C$$
$$\searrow D$$

$\{A, B, C, D\}$ bounded by B, D, k, and beam size

$\downarrow$

final chunk identities $\to$ layer-local full native K/V $\to$ transformer

Future Paper 3 interface: closure graph $\dashrightarrow$ progressive detail allocation

Figure 1: Iterative PRA increases search depth without transformer reruns. Gists are routing state; native K/V is the payload. The dashed handoff preserves graph signals for later work on partial materialization.

3 Method

3.1 Bounded frontier expansion

The root query initializes frontier $F_0 = \{q\}$ and the visited set is empty. At round t, each frontier vector proposes its k highest-scoring unseen chunks. Duplicate proposals are merged, the best path to each candidate is retained, and at most the remaining beam and unique-node budgets are accepted. Accepted gists form F_t. Traversal stops at fixed depth D, final budget B, no-new-node convergence, or an optional confidence threshold.

To limit semantic drift, candidate c is scored against both the root and its frontier parent:

$$e_t(c) = \alpha s(q, g_c) + (1 - \alpha) s(f_{t-1}, g_c), \qquad 0 \le \alpha \le 1. \tag{3}$$

At $\alpha = 0$, traversal follows local gist edges. At $\alpha = 1$, every round reproduces root ranking after visited-node removal. Intermediate values preserve the original information need while allowing a retrieved node to redirect search.

The primary path score multiplies affinities $a_i = (e_i + 1)/2$ along a path:

$$P(c_t) = \prod_{i=1}^{t} a_i. \tag{4}$$

We also evaluate log-sum, final-edge, minimum-edge, mean-edge, and direct-root scores. These choices are deliberately transparent. A learned controller would confound the first question: whether useful graph structure is already present.

Listing 1: Level-1 bounded associative closure

```
frontier = {root}; selected = {}; graph = empty
for hop in 1..D:
    proposals = union topk_unseen(parent, G, k) for parent in frontier
    merge duplicate targets; retain parent edges and best path
    accepted = top(proposals, min(beam, B - len(selected)))
    add accepted nodes and edges to graph
    frontier = winning_gists(accepted)
    stop if frontier is empty or len(selected) == B
materialize_full_native_kv(selected) # only after closure
```

3.2 Level-2 frontier updates

Direct Level-1 traversal uses each winning gist as the next query. We compare three parameter-free updates. A residual update keeps parent state,

$$r_{t+1} = \text{normalize}(r_t + \beta g_t), \tag{5}$$

while mean and path-weighted mean collapse accepted branches to one aggregate vector and add the root. These updates test whether mild state accumulation reduces drift. They do not rerun a transformer and introduce no learned parameters.

3.3 Asymmetric projection contract

Paper 2's frozen router does not embed questions and memories with one shared map. It computes

$$z_q = W_q h_q, \qquad z_m = W_m h_m, \tag{6}$$

and scores $s(q, m) = \cos(z_q, z_m)$. The original closure stored z_m and reused it as the next frontier. Its second hop was $\cos(W_m h_A, W_m h_B)$, although the learned interface expects

$$\boxed{s(A, B) = \cos(W_q h_A, W_m h_B).} \tag{7}$$

Equal tensor width does not make the two projections semantically interchangeable. The corrected cache derives and stores an aligned query-projected companion gist when it builds the memory index. No parameter changes and no retraining are involved. Root ranking remains $W_q h_q \to W_m h_B$; only hop 2 and later use $W_q h_A$.

3.4 Contextual parents and local propagation

Chunk means can hide a narrow relational bridge. We therefore distinguish three scales:

$$\boxed{\text{encoding granularity} \neq \text{propagation granularity} \neq \text{materialization granularity.}} \tag{8}$$

The fixed baseline encodes a 256-token parent once through Qwen. It divides that already contextualized hidden-state tensor into eight contiguous 32-token windows and mean-pools one local gist per window. Local windows are never independently re-encoded. A node has identity (parent_id, local_id) and carries its source span and both W_q and W_m projections. Traversal is

$$q \to g_{A,i} \to g_{B,j}, \qquad s(A_i, B_j) = \cos(W_q h_{A,i}, W_m h_{B,j}). \tag{9}$$

Candidate local nodes are aggregated by parent, previously selected parents are removed, and the best local activation proposes each new parent at most once. Final selection materializes whole 256-token parent K/V under the same parent-count budget as one-shot and parent closure. The experiment records local nodes explored, cross-parent transitions, local and parent entropy, and

$$\Delta_{bridge} = s(g_{A,i}, g_{B,j}) - s(g_A, g_B). \tag{10}$$

3.5 Budgets, cycles, and layer locality

Four controls bound expansion: depth D, neighbors per frontier node k, per-round beam size, and maximum unique chunks B. A visited set prevents $A \to B \to A$ from consuming budget twice. Alternate parents remain graph edges even when one path wins pruning. Every index is layer-local. A query, gist, and native payload from different layers are never treated as interchangeable merely because their dimensions match.

If C is the number of chunks and $|F_t|$ the frontier size, Level-1 scoring performs $C \sum_t |F_t|$ gist comparisons. Full K/V transfer remains proportional to the final selected payload, exactly as in matched one-shot PRA. The method trades more compact-index work for a chance to find a better closure without increasing active K/V.

4 Experimental Design

4.1 Inherited Paper 2 configuration

Table 1 makes the inherited stack explicit. Layer numbers are zero based. The retrieval studies operate on frozen feature artifacts and do not run the transformer or activate K/V; the separate downstream smoke is the only Paper 2.5 experiment that exercises the complete native-K/V consumer path.

Table 1: Verified inherited configuration and Paper 2.5 experimental scope.

Component	Paper 2.5 setting
Backbone	`Qwen/Qwen3-0.6B`, frozen; 596,049,920 parameters and no trainable backbone parameter during feature capture.
Backbone revision	Hugging Face commit `c1899de289a04d12100db37 0d81485cdf75e47ca`. The inherited feature manifest and downstream smoke pin this hash. The Gate-2 manifest recorded the alias `main`; the cached ref used for that run resolves to the same hash.
Backbone topology	28 decoder layers (zero-based 0–27), hidden width 1,024, 16 query heads, 8 K/V heads, and native head width 128.
Routing source	Attention-input hidden states from decoder layer 27; the root uses the final prompt-token hidden state.
Contextual encoding and routing unit	The inherited feature artifact encodes contiguous 128-token blocks and exposes 32-token routing views. These views are derived from contextual hidden states, not independently encoded fragments. Gate 2 separately encodes 256-token parents once and derives eight 32-token local means from each parent state.
Raw routing gist	One mean hidden-state gist per inherited 32-token routing chunk. Gate 2 retains one local mean per 32-token subwindow and an occupancy-weighted 256-token parent mean.
Native closure representation	Gate 3 uses tokenwise layer-27 pre-RoPE Q/K inside contextual 32-token regions, with Qwen-compatible 16-query-head to 8-K/V-head pairing. These tensors are routing only; the materialized payload remains post-RoPE native K plus native V.
Learned discovery component	The only inherited learned discovery component is Paper 2's asymmetric linear semantic router. It maps a 1,024-D hidden state to a 128-D routing space with 262,144 parameters.
Router projections	$z_q = W_q h_q$ and $z_m = W_m h_m$; one-shot scoring is $\cos(z_q, z_m)$.
Router checkpoints	Five independently trained exhaustive-negative, margin-objective checkpoints with seeds 11, 23, 37, 53, and 71. No checkpoint is trained or selected on Paper 2.5 iterative results.

Table 1 (continued).

Component	Paper 2.5 setting
Iterative closure	Parameter-free. The primary chunk study uses direct frontiers, multiplicative path score, root anchor $\alpha = 0.25$, and depths 2, 3, and 4; one-shot is the matched depth-1 Top-B reference. Gate 3 also remains parameter-free.
Memory-consumption adaptation	None. Paper 2.5 does not load a memory-use adapter. Residual-16, rank-32 LoRA, residual-plus-LoRA, and global LM-head adaptation are all absent.
PRA payload	Full layer-native post-RoPE K and native V, preserving Qwen's 8 K/V heads without expanding them to 16 query heads.
PRA consumer layers	None in the frozen-feature retrieval experiments. The downstream integration smoke routes at layer 27 and materializes/consumes native K/V on layer 27 only; it does not use a multi-layer late band.
Routing versus materialization	Intermediate closure nodes remain compact routing state and never enter attention. Only final selected identities are handed to the existing layer mapper before native K/V becomes active.
Native-operation bounds	The configured hard bound is 256 tokens. Feature encoding uses 128-token blocks in the inherited study and 256-token parents in Gate 2. The downstream smoke uses 64-token encoding blocks, observes a maximum native operation of 64 tokens, and records zero violations.
Final native payload budget	The downstream smoke caps materialized memory at 96 tokens and selects three 32-token chunks in both one-shot and iterative conditions. Retrieval-only studies apply matched logical chunk/parent budgets but do not transfer K/V.
K/V residency in smoke	Reference K/V is CPU-resident, pinned when CUDA is available, and transferred non-blockingly for final selected identities.
Datasets	A held-out split of 16 HotpotQA and 16 QASPER examples, plus controlled synthetic two-hop, three-hop, and local-bridge tasks.
New Paper 2.5 variables	Discovery procedure only: one-shot versus bounded iterative closure, followed by projection-correct frontier, parent-versus-local propagation, and semantically narrowed native-QK propagation gates. Backbone weights, router checkpoints, payload representation, layer mapper, and host attention semantics remain fixed within each matched comparison.

The resulting path is

frozen Qwen hidden state → learned 128-D asymmetric router → one-shot or iterative closure → final identities → PRA layer mapper → native K/V on layer 27.

Iterative closure operates only over compact routing state. Intermediate graph nodes do not inject K/V into the transformer. After closure terminates, the final selected identities are handed to the existing PRA layer mapper, which materializes the corresponding full native K/V payload on the configured consumer layer in the downstream smoke.

Paper 2.5 changes the discovery procedure only. The pretrained backbone, routing checkpoints, native-K/V payload representation, layer mapper, and host-model attention semantics remain fixed within each comparison. Therefore differences between one-shot and iterative conditions are routing/closure effects, not effects of additional model training or a different memory-consumption adapter.

Table 2: Isolation of inherited components from Paper 2.5 contributions.

Inherited from Paper 2	New in Paper 2.5
Frozen pinned Qwen3-0.6B	Bounded iterative closure and retrieval graph
128-D asymmetric semantic router	Depth, path, root-anchor, and stop logic
Attention-input hidden-state gists	Projection-correct frontier audit
Native post-RoPE K/V transport and mapper	Hierarchical local propagation with parent deduplication
Bounded native execution	Matched-budget chain metrics and transitive controls
Fixed post-RoPE native K/V payload	Pre-RoPE native-QK propagation audit

4.2 Frozen semantic geometry

We reuse Paper 2's frozen Qwen3-0.6B feature artifact and its attention-input hidden-state representation [5, 9]. The held-out split contains 16 HotpotQA and 16 QASPER examples, 32-token chunks, exact evidence-overlap masks, and 1,024-dimensional root and memory features [11, 2]. Five asymmetric linear projections, each with 262,144 parameters and seeds 11, 23, 37, 53, and 71, map query and memory features to 128 dimensions. They were trained on the disjoint Paper 2 train split with exhaustive negatives. No router is trained or selected on iterative results.

The primary setting uses direct frontier gists, multiplicative path score, $\alpha = 0.25$, and $D \in \{2, 3, 4\}$. Final budgets are 5, 10, 20, and 30% of each example's candidate chunks. One-shot always receives the identical final B. We separately sweep $\alpha \in \{0, .25, .5, .75, 1\}$, all Level-2 updates, and all six path scores at $D = 2$, 10%.

4.3 What the evidence metrics mean

Any-evidence recall asks whether one annotated chunk is selected. Exact all-evidence recall asks whether every evidence-overlapping 32-token chunk is selected. Because one supporting sentence can cross a chunk boundary, we additionally group contiguous positive chunks and define chain completion as selecting at least one chunk from every group. The artifact reports budget feasibility for both definitions. Hotpot groups are ordered by source position; they are not claimed to encode causal hop order. QASPER usually has one contiguous evidence group and is therefore a direct-retrieval control.

The synthetic controls provide exact causal chains. Their root is similar to node A; each next node is similar to its predecessor but is outranked from the root by distractors. At matched $B = 2$ or 3, one-shot can afford the entry node but not the full chain. We run 64 examples per seed for each chain length, yielding 320 paired trials per task.

4.4 Downstream execution validation

A separate two-example CUDA smoke loads the pinned Qwen model, indexes one HotpotQA and one QASPER source, and compares no memory, one-shot, and $D = 2$ closure. One-shot and closure both request three chunks and materialize 96 native K/V tokens. This run validates execution and budget invariants; two examples cannot estimate answer quality.

4.5 Gate-3 native-QK protocol

Gate 3 asks whether initial relevance and associative propagation require different pretrained geometries. The learned semantic router still scores the root and narrows each activated local frontier to either 10% or 20% of candidate parents. Only local regions inside those parents enter native refinement. From the same layer-27 contextual states used by Gate 2, we retain Qwen's normalized pre-RoPE projections

$$Q^{pre} = W_Q h, \qquad K^{pre} = W_K h. \tag{11}$$

Qwen has 16 query heads and 8 K/V heads, so query heads $2j$ and $2j+1$ compare only with K/V head j. The primary local-region transition score is

$$s_{QK}(A_i, B_j) = \max_{a \in A_i, b \in B_j, h} \frac{q_{a,h}^{\top} k_{b,\lfloor h/2 \rfloor}}{\sqrt{128}}. \tag{12}$$

The sole smooth control averages the four strongest valid token pairs per head and then the four strongest query heads. We compare fixed Top-k transitions with one threshold, $\theta = \mu + \sigma$, always under the same hard parent cap. If thresholding accepts too few parents, root-semantic ranking fills the remainder so final materialization stays matched.

Each 256-token parent is encoded once. Its token states are sliced into contextual 32-token routing regions; no region is independently re-encoded. Native QK changes discovery identities only. Intermediate states contain no payload K/V, and final identities would still map through the existing post-RoPE native K plus native V path. The five router seeds, 10/20/30% final parent budgets, and all one-shot, parent-closure, and local-semantic baselines remain fixed. These Gate-3 baselines are recaptured on the same strict 32-token grid as native QK. They are internally matched but slightly differ from Gate 2's historical eight-equal-segment artifact when a final 256-token parent is short; Tables 6 and 7 therefore answer their respective within-gate comparisons rather than being pooled.

5 Results

5.1 Exact latent chains validate the mechanism

Table 3 gives the cleanest mechanism test. Both methods find some evidence, but one-shot retrieves only the entry node. Iterative traversal follows the designed edges and completes every chain. The gain costs one complete index scan per hop.

Table 3: Matched-budget synthetic results over five seeds and 64 examples per seed.

Task	Method	Any	Exact chain	Coverage	Comparisons
2-hop	One-shot Top-2	1.000	0.000	0.500	10
2-hop	Iterative $D = 2$	1.000	1.000	1.000	20
3-hop	One-shot Top-3	1.000	0.000	0.333	11
3-hop	Iterative $D = 3$	1.000	1.000	1.000	33

This result establishes that the implementation can recover indirect nodes without transformer updates. It does not show that a pretrained semantic router supplies the required edges.

5.2 HotpotQA shows a small chain signal and a larger drift cost

Figure 2 and Table 4 compare natural-data routing. At 10%, one-shot any-evidence recall is 0.700 ± 0.065 (five-seed 95% interval), while $D = 2$ obtains 0.663 ± 0.118. Yet group-level chain completion rises from 0.138 to 0.150, and reaches 0.175 at $D = 4$. The two metrics describe different behavior: closure sometimes reaches another evidence group, but also leaves an easily retrieved entry neighborhood.

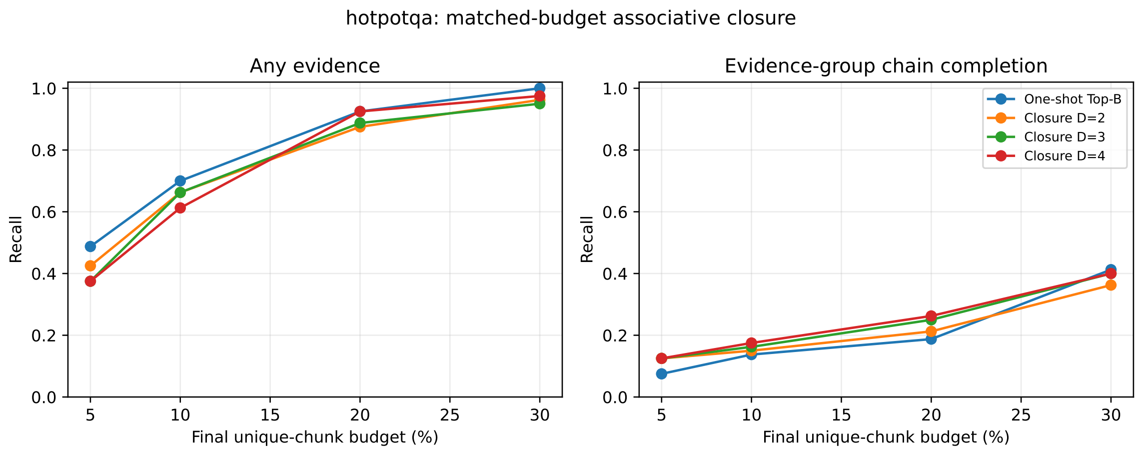

Figure 2: Five-seed HotpotQA matched-budget curves. Deeper closure modestly improves chain completion at some sparse budgets but does not improve the broader any-evidence curve.

Table 4: Primary 10% matched-budget results. Means combine 16 examples and five independently trained routers. "All" requires every positive chunk; "Chain" requires every evidence group.

Dataset	Method	Any	Chain	All	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	Closure $D = 2$	.663	.150	.000	.202	219.6
HotpotQA	Closure $D = 3$	.663	.163	.000	.205	273.0
HotpotQA	Closure $D = 4$	.613	.175	.000	.193	303.2
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	Closure $D = 2$	.875	.875	.200	.466	1395.9
QASPER	Closure $D = 3$	.788	.788	.175	.425	1822.1
QASPER	Closure $D = 4$	.825	.825	.150	.426	2144.2

At 20%, Hotpot chain completion rises from 0.188 one-shot to 0.212, 0.250, and 0.263 at depths 2–4. At 30%, one-shot returns to the lead (0.412 versus 0.362–0.400). The effect is therefore small, budget dependent, and not accompanied by improved exact all-chunk recall. Five seeds are insufficient to claim a reliable improvement; the direction also changes with budget.

5.3 QASPER behaves as a direct-retrieval control

QASPER's evidence commonly forms one contiguous group, so chain completion equals any-evidence recall in this protocol. Figure 3 shows that one-shot dominates throughout the sparse region. At 10%, its any/chain recall is 0.963 ± 0.069 versus 0.875 ± 0.095 for $D = 2$. Closure adds no useful intermediate relation when the root already addresses the target.

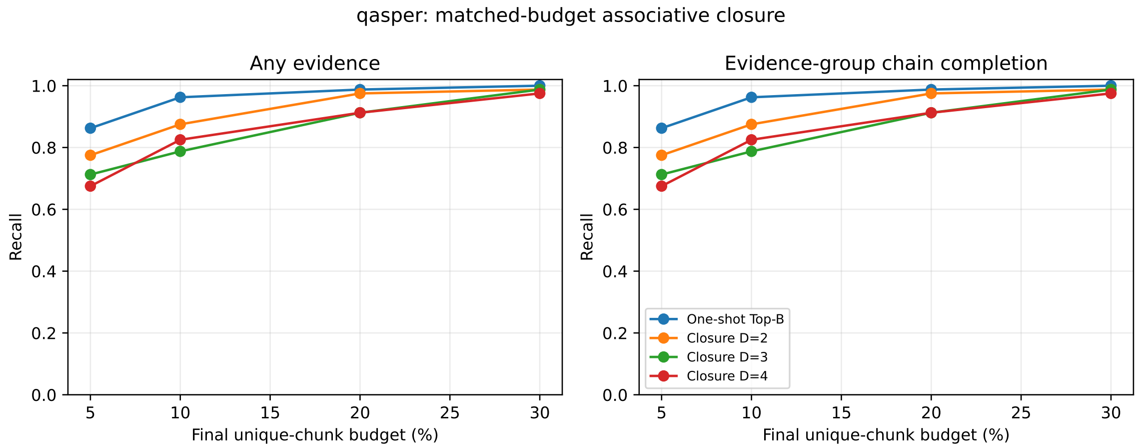

Figure 3: QASPER matched-budget curves. The mostly single-group evidence protocol favors direct one-shot ranking, as expected if recursion is unnecessary.

5.4 Root anchoring exposes semantic drift

At 10% Hotpot, pure local traversal ($\alpha = 0$) yields any recall 0.600. Increasing root weight to 0.50, 0.75, and 1.0 gives 0.700, 0.738, and 0.700. QASPER improves monotonically toward the root-only condition: 0.863, 0.875, 0.900, 0.913, and 0.963 from $\alpha = 0$ to 1. The best QASPER "iterative" setting is therefore the one that does not use frontier semantics.

Residual update is the only Level-2 variant to preserve exact all-chunk recall on Hotpot (0.013), but its any recall 0.713 does not consistently beat one-shot. Mean and weighted mean reach 0.738 any recall while exact all remains zero. Path-score variants range from 0.650 to 0.738 Hotpot any recall and do not improve chain completion consistently. No parameter-free update dominates.

5.5 Relational gap does not identify real gains

The motivating diagnostic is

$$\Delta_{rel} = \max_{A \in E \setminus \{B\}} s(g_A, g_B) - s(q, g_B). \tag{13}$$

Large Δ_{rel} should favor closure if evidence-to-evidence similarity is a useful edge. Figure 4 does not show that pattern. In Hotpot's highest gap quartile, the paired $D = 2$ any-recall effect is -0.100 and coverage effect is $+0.017$. In QASPER they are -0.176 and -0.019. A high pairwise cosine score is not enough to identify the correct relational transition.

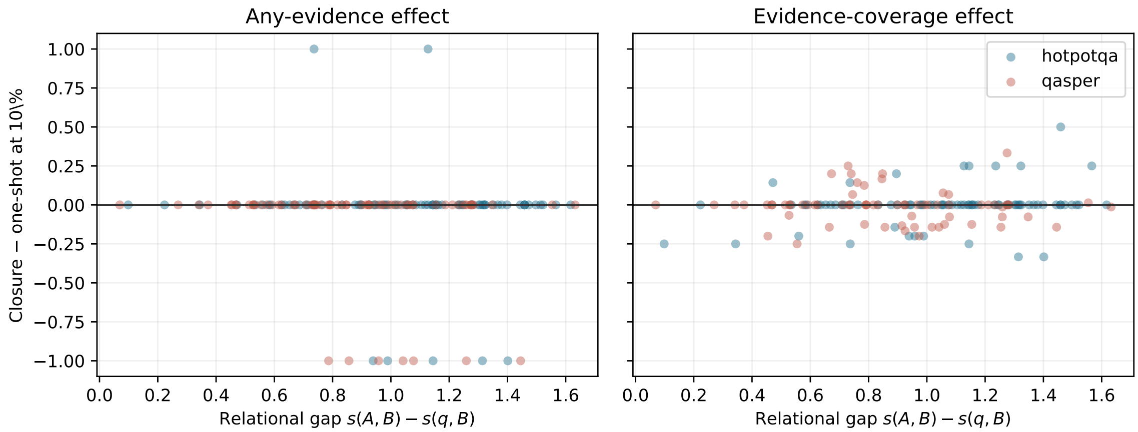

Figure 4: Paired $D = 2$ minus one-shot effects at 10%. Relational gap does not predict a stable closure advantage in the inherited semantic space.

5.6 Search cost and full-K/V execution

Figure 5 shows the compact-index cost. At 10%, $D = 2$ uses 219.6 comparisons per Hotpot example versus 50.1 one-shot (4.4×), and 1395.9 versus 142.6 on QASPER (9.8×). The higher QASPER multiplier follows from more candidates and larger frontiers. These are exact comparison counts; wall time depends strongly on batching and hardware.

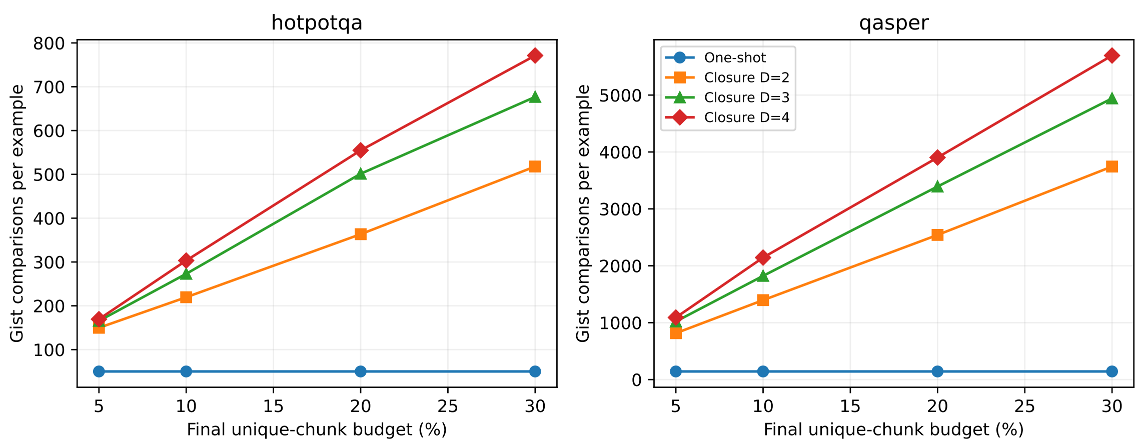

Figure 5: Compact gist comparisons. Iteration leaves final K/V budgets fixed but increases index work with frontier size and depth.

The CUDA integration smoke materializes exactly three chunks and 96 K/V tokens for both methods. Iterative routing takes 0.103/0.077 seconds on the Hotpot/QASPER examples, versus 0.107/0.073 for one-shot in this trace-oriented implementation. End-to-end times are 1.291/1.291 seconds for closure and 1.304/1.347 seconds for one-shot. The model records zero native-limit violations and a maximum 64-token native operation. These two cases validate the execution contract only.

6 Projection and Locality Gates

The preceding results freeze the original chunk-level closure as Level 1. We next change one assumption at a time. All comparisons reuse the same five checkpoints, held-out examples, depth, root anchor, path score, and final unique-parent/K/V budget. Neither gate trains a new parameter.

6.1 Gate 1: projection-correct frontiers

Table 5 compares the historical $W_m h_A$ frontier with the contract-correct $W_q h_A$ frontier at the primary 10% chunk budget. Correction materially reduces the QASPER failure: any/chain recall rises by 0.075 and coverage by 0.013 relative to historical closure. It nevertheless remains 0.013 below one-shot any/chain recall. On HotpotQA, correction raises any recall by 0.025 and restores exact identity by 0.013 relative to historical closure, but chain completion falls by 0.013 and exactly matches one-shot. Comparison counts are unchanged because this gate changes only the projection used by each activated frontier.

Table 5: Gate 1 at a matched 10% final chunk budget. Q-frontier is the corrected $W_q h_A \to W_m h_B$ traversal; M-frontier is the historical direct reuse of $W_m h_A$.

Dataset	Method	Any	Chain	Exact	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	M-frontier	.663	.150	.000	.202	219.6
HotpotQA	Q-frontier	.688	.138	.013	.201	219.6
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	M-frontier	.875	.875	.200	.466	1395.9
QASPER	Q-frontier	.950	.950	.225	.479	1395.9

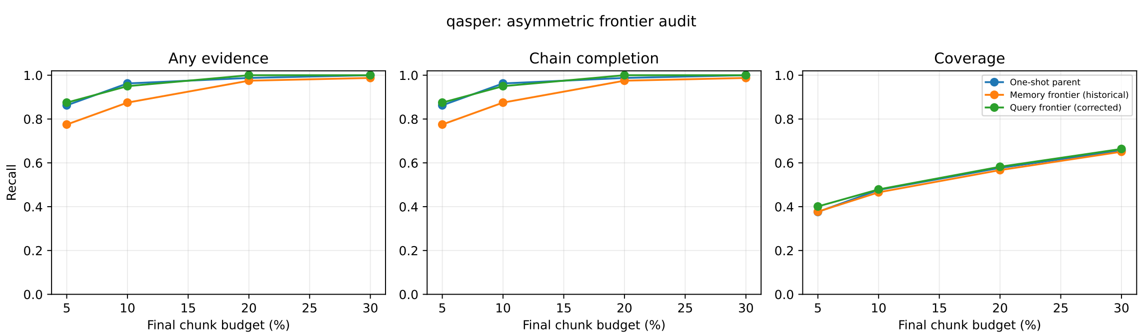

Figure 6: Gate-1 QASPER curves. Respecting the asymmetric projection contract recovers most of the historical closure loss, but one-shot remains the sparse-retrieval reference.

Projection asymmetry therefore explains a meaningful part of the original degradation. It does not explain why the corrected traversal fails to improve chain completion. That result passes the predeclared checkpoint for testing a finer propagation scale.

6.2 Gate 2: contextual local gists

Gate 2 changes the index to 256-token contextual parents with eight 32-token local gists. The one-shot and parent-closure baselines use the same parent features; local closure alone traverses subchunk states. Parent deduplication holds final materialization to the same parent count.

The designed local-bridge control separates the methods cleanly. At two selected parents, one-shot and projection-correct parent closure retrieve only the entry parent: exact identity, chain completion, and coverage are 0, 0, and 0.5 over 320 examples. Local closure retrieves both evidence groups in every case, with exact identity, chain completion, and coverage all 1.0. Its mean Δ_{bridge} is 1.0. The implementation can therefore expose a bridge erased by a parent mean without re-encoding the local window.

Natural data is more restrained (Table 6). At 20%, local closure improves Hotpot any recall, exact identity, and coverage by 0.013 each over one-shot, but chain completion falls by 0.013 and parent closure is stronger. On QASPER, local closure improves exact identity by 0.025 and coverage by 0.016 over one-shot while any/chain recall falls by 0.013. Local accepted edges are stronger than corresponding parent edges: mean Δ_{bridge} is 0.203 on Hotpot and 0.205 on QASPER. That measurable locality does not yet identify the correct cross-parent chain.

Table 6: Gate 2 at a matched 20% final parent/KV budget. Comparisons count semantic gist dot products; no native QK comparisons occur in this gate.

Dataset	Method	Any	Chain	Exact	Coverage	Gist cmp.	K/V tokens
HotpotQA	One-shot parent	.475	.138	.000	.217	6.5	405.8
HotpotQA	Parent closure	.513	.175	.000	.235	11.9	405.0
HotpotQA	Local closure	.488	.125	.013	.229	49.5	403.4
QASPER	One-shot parent	.938	.938	.738	.813	18.2	1036.0
QASPER	Parent closure	.950	.950	.738	.819	66.6	1037.3
QASPER	Local closure	.925	.925	.763	.829	405.2	1031.0

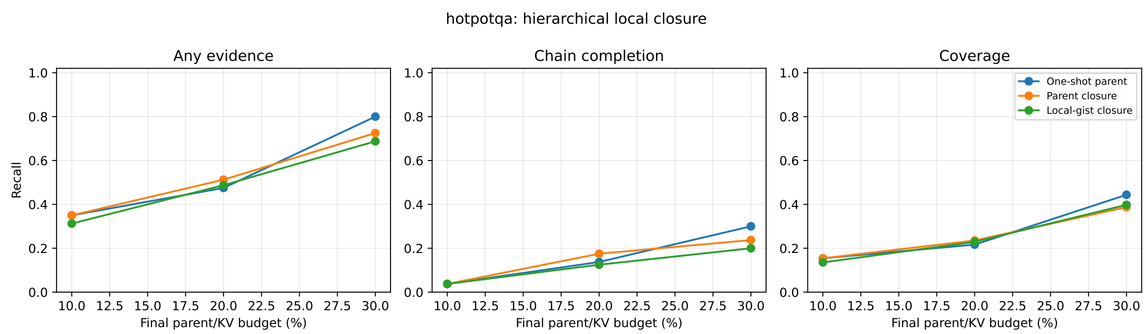

Figure 7: Gate-2 HotpotQA curves under matched final parent budgets. Local routing exposes stronger subchunk edges but does not improve evidence-group chain completion.

The cost is substantial. At 20%, local closure uses 49.5 versus 6.5 comparisons on Hotpot (7.6×) and 405.2 versus 18.2 on QASPER (22.3×). Materialized K/V differs by less than 1.3% within either dataset, confirming that the quality comparison is not purchased with a larger payload. The graph audit validates all 960 saved Gate-1/2 traces against schema v2 and finds no parent-budget violation.

6.3 Gate 3: native local QK propagation

The native-bridge control first verifies the mechanism without relying on natural annotations. The root-semantic rank retrieves entry parent A and a stronger semantic distractor, while a token in A has a model-compatible native QK match only to evidence parent B. Across five seeds and 64 cases per seed, one-shot, parent closure, and local semantic closure retrieve half the evidence and complete no chains. Native QK retrieves both groups in all 320 cases: exact identity, chain completion, and coverage are all 1.0. A distractor with an incompatible native key remains unselected. The vectorized scorer can therefore express the proposed transition.

Natural data does not reproduce that separation (Table 7). At the primary 20% final budget and 10% semantic candidate pool, max native QK matches local semantic chain completion on HotpotQA (0.125) and remains below one-shot (0.138) and parent closure (0.175). Its any-evidence recall and coverage are slightly higher than local semantic, but exact identity falls from 0.013 to zero. Expanding the pool to 20% lowers chain completion to 0.113. On QASPER, max native QK reaches 0.900 with the 10% pool and 0.863 with the 20% pool, below local semantic (0.925) and one-shot (0.938). More candidates amplify rather than repair propagation error.

Table 7: Gate 3 at a matched 20% final parent/KV budget. "QK-10" and "QK-20" are the primary max score with 10% and 20% semantic candidate-parent pools; "QK-Top4" is the sole Top-4 control at 20%. Native dots count token-pair/query-head products.

Dataset	Method	Chain	Coverage	Sem. cmp.	Native dots	K/V tok.	Time (s)
HotpotQA	One-shot	.138	.217	6.5	0	405.8	.0046
HotpotQA	Local semantic	.125	.217	98.3	0	407.0	.0105
HotpotQA	QK-10	.125	.223	114.6	95,712	406.3	.0121
HotpotQA	QK-20	.113	.208	114.6	183,667	407.2	.0132
HotpotQA	QK-Top4	.175	.265	114.6	183,667	406.0	.0143
QASPER	One-shot	.938	.813	18.2	0	1036.0	.0059
QASPER	Local semantic	.925	.829	540.0	0	1042.9	.0463
QASPER	QK-10	.900	.810	540.0	1,198,554	1043.7	.0467
QASPER	QK-20	.863	.794	540.0	1,949,958	1039.8	.0542
QASPER	QK-Top4	.838	.770	540.0	1,949,958	1041.3	.0624

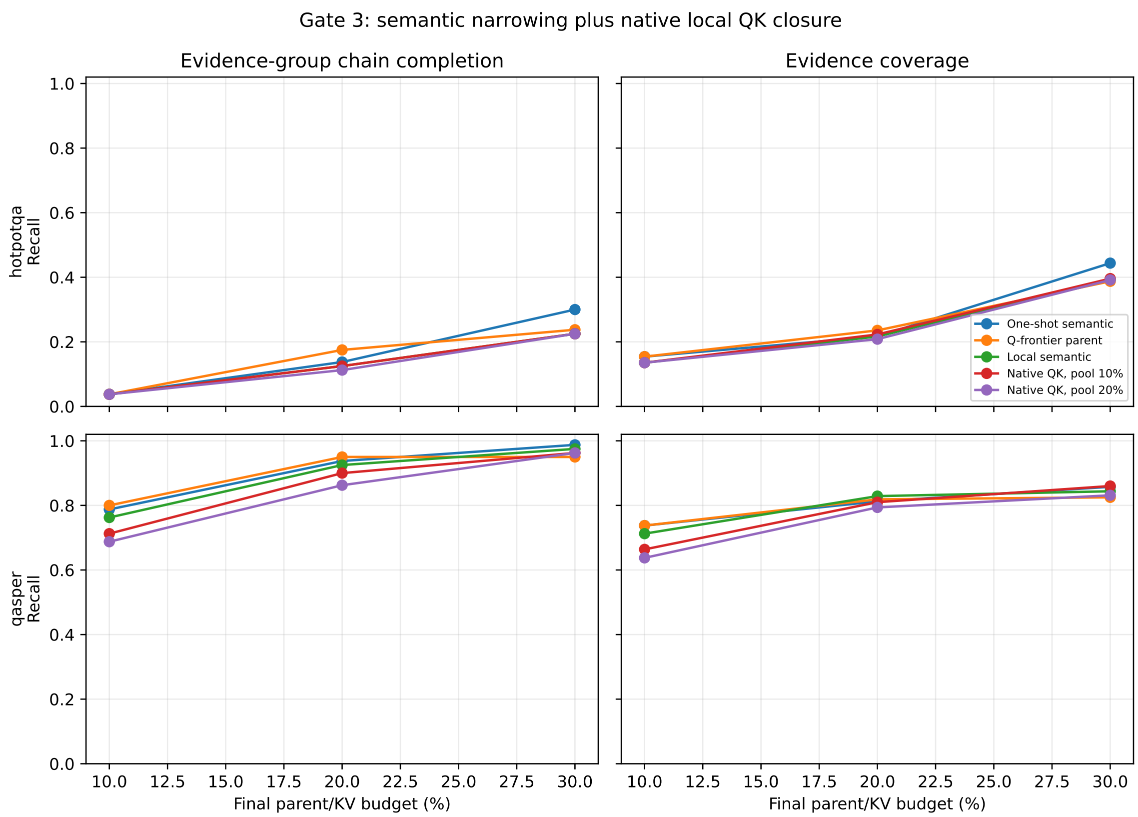

Figure 8: Gate-3 chain completion and evidence coverage across final budgets. Primary native-QK propagation does not dominate local semantic or one-shot routing; the larger candidate pool is usually worse, especially on the direct-retrieval QASPER control.

The Top-4 control is the one localized positive: HotpotQA chain completion at 20% rises to 0.175, matching parent closure and exceeding both one-shot and local semantic. It does not persist as a dominant curve: at 10% all methods tie at 0.038, at 30% Top-4 reaches 0.238 versus one-shot 0.300, and QASPER degrades to 0.838 at 20%. The $\mu + \sigma$ threshold is effectively identical to fixed Top-k in this hard-cap regime and offers no rescue. These predeclared controls are therefore a mixed point observation, not a positive Gate-3 result. This is specifically a failure of max-reduced Top-1 native closure under the tested competition rule; it does not yet show that useful native associative edges are absent. We test that distinction below.

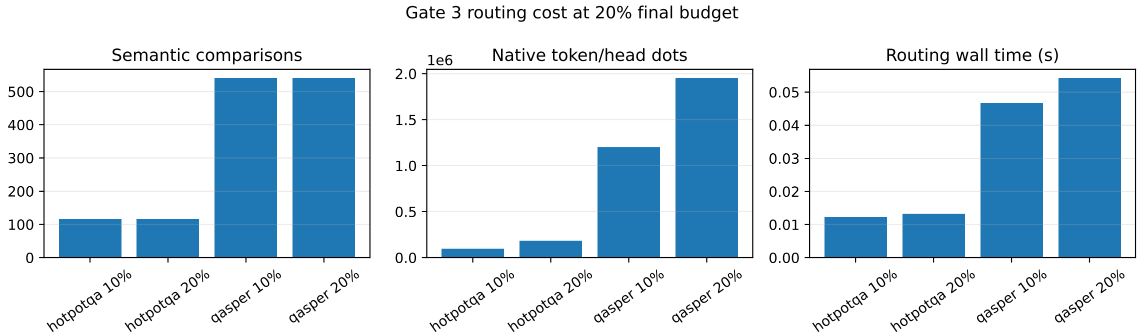

Figure 9: Gate-3 cost at the 20% final parent budget. Semantic and native work are reported separately. Candidate expansion from 10% to 20% nearly doubles native dot products without a quality gain. Wall time measures this batched reference implementation, not an optimized kernel.

At 20%, native refinement compares means of 1.1/1.7 candidate parents and 161/315 candidate tokens on HotpotQA for the 10/20% pools; QASPER compares 3.4/5.6 parents and 874/1,422 tokens. This produces 95,712–183,667 native dots on HotpotQA and 1.20–1.95 million on QASPER per example/seed, while materialized K/V stays within 0.8% of one-shot. Every saved graph reports zero K/V materializations during closure. Gate 3 changes search work and selected identities, not payload quantity or transformer attention semantics.

6.4 Oracle convergence and edge-rank diagnosis

The previous gates score whether a selected set touches each evidence group. Here we ask the stricter question of whether it approaches the exact feasible parent set used by the annotated Paper-2 oracle. We import that oracle constructor unchanged, pass it the cached evidence spans and 256-token parent boundaries, and assert equality with the feature mask for every example. For selected set $S(B)$ and oracle S_O, we report recall, precision, Jaccard overlap, and $\mathbf{1}[S_O \subseteq S(B)]$ at 5, 10, 20, 30, and 40% parent/KV budgets. Figure 10 shows that Hotpot convergence remains slow for every frozen method. At 20%, parent closure has the best non-control recall (0.235); the native Top-4 reduction reaches 0.265, but complete recovery is only 0.013. At 40%, one-shot is strongest at 0.529 recall, while all iterative variants remain between 0.438 and 0.477. Among the few Hotpot seed/example trajectories that ever recover the complete oracle, the mean minimum budget is 31.7–34.0%; only 10–12 of 80 trajectories do so by 40%.

QASPER provides the expected direct-retrieval control. At 20%, one-shot reaches 0.813 oracle recall and 0.738 complete recovery. Parent and local closure reach 0.819/0.738 and 0.829/0.763, respectively, but native max falls to 0.794/0.725 and the Top-4 reduction to 0.770/0.713. At 40%, one-shot and local differ by only 0.002 recall. Iteration therefore does not buy a useful convergence advantage on this mostly one-group task.

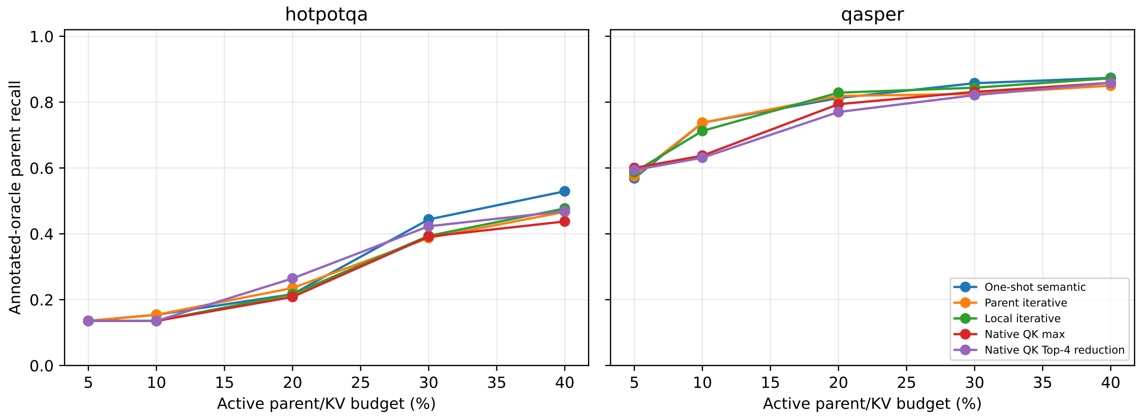

Figure 10: Exact Paper-2 oracle-parent recall versus active parent/KV budget. The Top-4 label denotes averaging the four strongest token/head matches, not a four-parent beam. HotpotQA improves slowly and remains root-limited; QASPER approaches its mostly direct oracle without requiring iterative native propagation.

Table 8: Oracle convergence at the primary 20% parent/KV budget. Semantic comparisons and native token-pair/head dot products are different accounting units and should not be added. Materialized K/V remains matched.

Dataset	Method	Recall	Precision	Jaccard	Complete	Sem. cmp.	Native dots
HotpotQA	One-shot	.217	.347	.171	.000	6.5	0
HotpotQA	Parent iterative	.235	.366	.183	.000	11.9	0
HotpotQA	Local iterative	.217	.353	.178	.013	98.3	0
HotpotQA	Native max	.208	.347	.169	.000	114.6	183,667
HotpotQA	Native Top-4 reduction	.265	.403	.211	.013	114.6	183,667
QASPER	One-shot	.813	.270	.259	.738	18.2	0
QASPER	Parent iterative	.819	.272	.260	.738	66.6	0
QASPER	Local iterative	.829	.271	.263	.763	540.0	0
QASPER	Native max	.794	.256	.250	.725	540.0	1,949,958
QASPER	Native Top-4 reduction	.770	.247	.241	.713	540.0	1,949,958

Low end-to-end recovery can arise before or after the first evidence group. At 20%, one-shot reaches the first source-ordered Hotpot group in 0.200 of seed/example runs; parent closure reaches 0.213, local closure 0.200, native max 0.188, and native Top-4 0.263. To isolate propagation, we then force only that annotated source group active and rank the next evidence group among all other parents. Any local node in either evidence parent may realize an edge. Parent semantic uses the projection-correct query-to-memory score, local semantic uses the strongest local pair, and native QK uses the exact frozen pre-RoPE max token/head reduction.

The conditioned result is much stronger than end-to-end closure (Figure 11). Median true-edge rank is 3 in all three spaces. Parent semantic, local semantic, and native QK respectively obtain R@1 of 0.141, 0.259, and 0.235; R@4 of 0.765, 0.741, and 0.765; and MRR of 0.424, 0.480, and 0.469. Every semantic transition is within Top-8, as are 16 of 17 native transitions. Mean oracle-minus-best-distractor margins remain negative ($-0.118, -0.143, -0.039$), so the correct edge is usually present but loses hard competition. Across 85 seed-transition comparisons, the best semantic and native ranks are both within Top-4 in 56 cases; semantic alone is near-top in 14, native alone in 9, and both are poor in only 6.

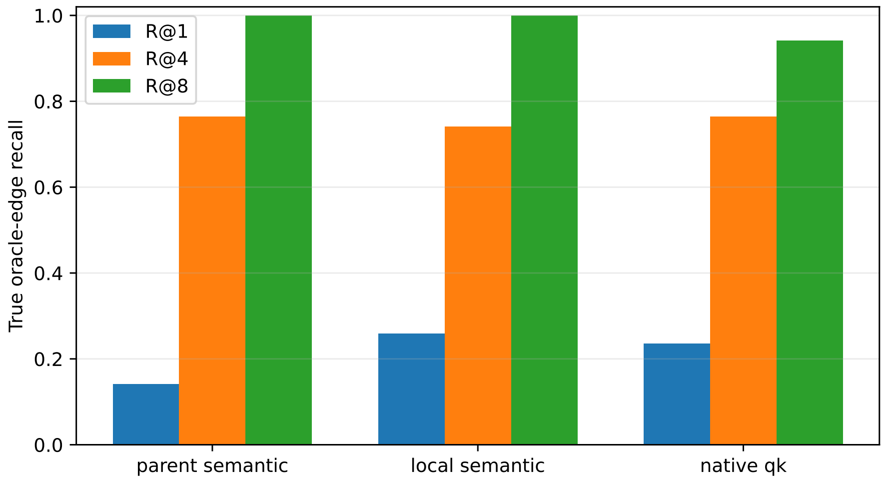

Figure 11: Rank recall for the true next Hotpot evidence group after conditioning on the first group. Most annotated transitions exist within the first four candidates even though they rarely win Top-1. Semantic statistics contain 17 transitions under five projection seeds; native QK has 17 seed-invariant transitions.

The Gate-3 Top-4 signal should not be read as parent-beam R@4. It averages four strong token/head matches before ranking parent transitions. Its four 20% chain wins over max reduction comprise two questions. In one seed-case a near-tied oracle edge moves from rank 2 to rank 1: its margin changes from -0.0038 to $+0.0157$. The other question accounts for three seed-cases, but its source-ordered edge remains rank 6 under both reductions; completion comes from root/reverse-order selection rather than better forward propagation. The cached features contain no entity/topic labels, so we do not assign a semantic relation to those distractors from parent IDs alone.

Finally, we simulate, without changing SDK routing, $k = 1$ when the Top-1/Top-2 gap exceeds a validation-selected threshold and $k = 4$ otherwise. On held-out edges, parent semantic selects $k = 4$ always and reaches 0.686; local semantic averages $k = 3.83$ and reaches 0.657 versus its 0.686 fixed-R@4 ceiling; native averages $k = 3.57$ and reaches 0.714, exactly its fixed R@4. The split is only 10/7 distinct native transitions and 50/35 seed-sensitive semantic transitions, so this is a mechanism diagnostic, not a tuned policy result. The simple margin rule saves little competition and does not explain the Top-4 reduction signal.

The decision is therefore *adaptive bounded competition next, not learned geometry yet.* True edges are usually near the top, which rejects the claim that both inherited spaces lack associative structure. A useful next rule must improve first-group discovery, admit several competitive transitions only when warranted, and beat fixed Top-4 cost on held-out examples. If calibrated competition fails that test, explicit edge supervision becomes justified.

7 Discussion

7.1 What the experiments establish

The synthetic result and the integration tests establish three mechanical facts. First, a gist can serve as a new query and recover a node that the root cannot afford at matched B. Second, deduplication, beam pruning, and hard budgets prevent exponential materialization. Third, the closure can remain entirely in compact routing state until final identities map to full native K/V.

The natural-data results establish a different boundary. Projection-correct closure confirms that shared dimensionality did not imply shared query and memory semantics: respecting the asymmetric interface repairs much of QASPER's loss. The local experiment then finds positive bridge deltas and better exact identity/coverage, so parent aggregation does hide some useful semantic relations. Yet those stronger local edges do not improve chain completion. Direct iterative reuse of the inherited one-shot routing representation does not yet provide a dependable associative traversal graph on natural data.

Gate 3 then tests the transformer's own position-neutral local matching geometry after semantic narrowing. Its exact synthetic recovery proves that the routing path, grouped heads, and tokenwise reduction can represent a native bridge. The natural result is nevertheless negative under the predeclared success criterion: primary native QK does not beat local semantic closure or one-shot across matched budgets. The isolated HotpotQA Top-4 gain is offset by other budgets and by QASPER degradation. Initial relevance geometry and propagation geometry may differ, but pretrained layer-27 QK does not establish that specialization here.

This completes the staged diagnostic sequence: Stage 0 is one-shot retrieval; Stage 1 is chunk iterative closure; Stage 2 corrects the asymmetric projection interface; Stage 3 moves propagation to contextual local semantic gists; Stage 4 tests tokenwise native QK after semantic narrowing. Projection role, propagation scale, and native attention geometry have therefore been tested before introducing learned associative supervision. Oracle-conditioned rank now shows that most true transitions survive within Top-4, so a learned head is not yet the next controlled intervention. Competition policy and root discovery must first be separated from representation failure; any later trained component should still be evaluated as new machinery rather than an automatic extension of the inherited router.

7.2 Relation to retrieval, agents, and recursive models

Memory Networks repeatedly read differentiable memory to support multi-hop tasks, while RAG, RETRO, and interleaved retrieval systems bring external evidence into generation [4, 1, 10]. Agentic systems can reformulate a text query after each model pass. Iterative PRA is narrower: it traverses already computed latent addresses inside one memory subsystem. It avoids repeated text serialization and full-model round trips, but it cannot replace an agent that must plan tools, verify evidence, or change the retrieval language.

Tiny Recursive Models repeatedly transform internal latent and answer state [3]. Controlled autoregressive work finds no reliable gain from the full autoregressive TRM architecture over simpler refinement baselines [6]. Iterative PRA does not apply a tiny neural network repeatedly. Its recurrence is *query* $\rightarrow$ *memory node* $\rightarrow$ *memory node*. This distinction motivates the parameter-free Level-1 test: retrieval recursion should earn its complexity before neural state recursion is introduced.

7.3 Paper 3 handoff

Papers 1–2 conservatively compress search but materialize every selected chunk in full. Iterative closure produces more structure before transfer: hop, parent edges, direct relevance, path score, confidence, branch entropy, and duplicate/overlap statistics. The versioned graph keeps these signals even when the final payload remains unchanged. Paper 3 can therefore ask a separate question: given a discovered closure, how much detail should each node receive? This paper does not test partial materialization.

8 Limitations

The held-out natural-data sample has 16 examples per dataset, repeated over five router seeds. Seeds quantify projection sensitivity but do not create 80 independent questions. Hotpot evidence groups are source-contiguous spans, not annotated causal hop labels. QASPER's selected subset is yes/no and often one-group, so it tests direct retrieval more than multi-hop closure.

The natural-data sample remains small. Projection correction, local hierarchy, and native QK remove three specific confounds, but the experiment does not include overlapping local windows, adaptive zoom rules, evidence-edge training, a layer sweep, or a full agentic baseline. The local and native bridge controls are deliberately geometric and do not approximate natural language. Hotpot evidence groups are approximate source-order targets rather than annotated transition edges, so they can understate or misidentify the useful successor relation. Downstream F1/EM is measured only in a two-example execution smoke and is not used as evidence of quality. Finally, the scorer batches token pairs and heads but still launches small per-example routing operations; the reported wall time is not a production-kernel claim.

9 Conclusion

Bounded associative closure is exact when latent memory contains the right edges. Correcting the asymmetric frontier is necessary and materially reduces inherited degradation. Routing at a finer local scale can recover bridges erased by parent means. Tokenwise pre-RoPE native QK then recovers every designed native bridge, showing that semantic narrowing and a different propagation primitive can compose without changing materialization. None of these parameter-free variants, however, produces a dependable natural-data chain advantage at matched payload budget, and each finer search adds work. One-shot PRA therefore remains the default efficiency/quality reference.

The scientific lesson is narrower than "pretrained geometry has no associative edges." Direct iterative reuse is sensitive to projection interface, propagation scale, candidate width, and score reduction. Local semantic and native attention spaces contain designed bridges and occasional natural positives, but neither supplies reliable cross-parent transitions on this sample. The oracle-rank diagnosis narrows the next step further: useful transitions are commonly present below Top-1, while root evidence discovery remains weak. Bounded adaptive competition should be tested before learned geometry, although the simple score-gap policy reported here mostly reduces to fixed Top-4. If a calibrated policy cannot convert near-top edges into held-out closure gains, future learned associative geometry must demonstrate that supervision adds stable transition structure beyond all four frozen discovery stages while retaining the same payload and execution controls.

A Implementation Map

The implementation is designed so another system can reproduce the mechanism without adopting the entire PRA-HF package. File and line references below refer to the accompanying source revision.

Layer-local packed index. `src/pra_hf/iterative.py:69--134` packs variable gist sets as `[chunks,max_gists,routing_width]` plus a validity mask. Records retain chunk identities and payload handles, but packing copies no native K/V.

Asymmetric companions and contextual parents. Lines 387–438 of `injection.py` project each contextual local gist with both W_m and W_q and projects the occupancy-weighted parent hidden-state mean separately. The full path is `src/pra_torch/hf/injection.py`. The cache stores compact companion vectors, not another native K/V payload.

Hierarchical index and parent deduplication. Lines 137–254 of `iterative.py` define aligned parent/local tensors and construct them from contextual segment gists. Lines 641–787 perform root-to-parent/local and local-to-local expansion, aggregates candidates by parent, enforces the unique-parent budget, and maps only final parents to payload handles.

Graph contract. Lines 269–342 of `iterative.py` define node, edge, graph, and result records. The current JSON Schema is `docs/papers/shared/schemas/retrieval_graph_v2.schema.json`. Nodes retain parent identity, local span, resolution, representation and projection types, hop, path score, and materialization state. Native-edge extensions additionally retain source/target spans, query and K/V head identities, native score, threshold, and semantic candidate rank.

Exact scoring and deterministic top-k. `iterative.py:376--397` normalizes frontier rows and computes `einsum("fd,cgd->fcg")` before max reduction across gists. A stable index tie-break is applied before `torch.topk`, producing deterministic identities on CPU and CUDA.

Traversal. `iterative.py:399--614` performs root scoring, frontier expansion, anchoring, deduplication, path pruning, Level-2 updates, stopping, and cost accounting. The code never accesses `chunk.token_kv`. Conversion at lines 617–638 returns selected payload handles only after the graph is final.

Host-model integration. `src/pra_hf/model.py:374--439` captures one root query with memory disabled, dispatches one-shot, chunk iterative, or opt-in `local_iterative` routing, maps final parent identities to consumption layers, enables fixed memory, and only then marks graph nodes materialized. Generation thereafter uses the unchanged Paper 2 native-K/V path.

Native QK feature capture. `precompute_native_qk_features.py:50--107` encodes each bounded 256-token parent once, reads Qwen's exact layer-27 capture, transposes canonical Q/K to token-major form, and slices aligned 32-token regions. It stores FP16 pre-RoPE Q/K with a validity mask for a short final window; it never independently re-encodes a local region. The capture itself is generated by Qwen's ordinary `q_proj`/`k_proj` and native Q/K norms before rotary position encoding. A test recomputes those projections from the captured hidden states and requires exact tensor equality.

GQA-compatible tokenwise scoring. Lines 122–211 of `src/pra_hf/native_closure.py` map query head h to K/V head $\lfloor h/2 \rfloor$, packs candidate keys accordingly, and computes all frontier-by-candidate token/head products with one `einsum`. Invalid padded pairs become $-\infty$ before max or Top-4 reduction. The scorer returns strongest token and compatible-head identities as well as the physical dot-product count; no Python loop traverses token pairs or heads.

Semantic narrowing and native propagation. `native_closure.py:215--480` uses root semantic scores for initial parents, applies $W_q h_A \to W_m h_B$ to rank a hard 10/20% candidate-parent pool, gathers all contextual local windows in that pool, and invokes native QK once. Proposals are merged by parent before the hard final parent budget. Threshold mode retains the same cap and fills rejected slots from root semantic ranking, preserving materialization parity. The graph cost record separates semantic comparisons, native dots, candidate parents/regions/tokens, accepted transitions, selected parents, final K/V tokens, and routing wall time.

Oracle and edge-rank diagnostic. Lines 66–116 of `run_oracle_convergence.py` map each annotation to overlapping parents and call the Paper-2 oracle constructor unchanged; disagreement with the cached evidence mask is fatal. Lines 133–185 define target rank using only frozen score vectors, with equal scores sharing a rank and the forced source group excluded. Parent/local/native score vectors are constructed at lines 188–245; target identity enters only the subsequent metric. Lines 252–292 fit the offline margin threshold on a hash-stable validation partition and evaluate it unchanged on test. The main loop beginning at line 588 wraps the existing Gate-2/3 evaluators, so convergence does not duplicate or subtly alter their selection policy.

Fixed-identity payload parity. `tests/test_hf_integration.py` sends the same selected chunk through router, oracle, and native-closure labels. All three paths produce byte-identical post-RoPE K, native V, and attention output. Gate 3 therefore ends at identities: it neither changes nor reconstructs the payload consumed by Qwen.

Reproduction. The experiment directory is `experiments/paper2_5_iterative_pra/`. Its Gate-1 runner isolates projection correction. Gate 2 captures one 256-token contextual parent pass and derives 32-token local means. Gate 3 uses `precompute_native_qk_features.py` followed by `run_gate3_native_qk_closure.py`. The 620,984,671-byte feature cache is regenerable and excluded from Git; its public manifest pins Qwen revision `c1899de289a04d12100db37 0d81485cdf75e47ca`. Its SHA-256 is

`edc589cc7886ba843ee598e89e6dab97e500084df8b7ee8dc8ab44ea75993ae8`.

Exact commands are listed in the experiment `README.md`; raw rows, aggregate tables, plots, and graph JSONL are under the corresponding shared result tree.

B Validation Gates

The test suite checks $D = 1$ equivalence to Top-B, deterministic batches, exact synthetic two-hop recovery, deduplication and cycles, depth/beam/unique budgets, all path and frontier modes, zero limits, layer locality, projection-contract enforcement, exact local-bridge recovery, parent deduplication, GQA head mapping, max and Top-4 native reductions, semantic narrowing, threshold caps, exact pre-RoPE Q/K capture, CPU/CUDA identity parity, fixed-identity payload parity, and the no-materialization-before-closure boundary. Gate-3 generation validates every saved graph against the backward-compatible schema-v2 extension and records zero K/V materializations during closure. The feature manifest verifies the frozen model revision, layer 27, 16/8 grouped heads, native head width 128, and no memory-use adapter. The complete repository suite passes 385 tests; the new focused oracle/native suite passes 16. All 160 saved Gate-3 graphs and 5,600 result rows satisfy schema, parent-budget, and no-closure-materialization checks. The additive diagnostic preserves 4,000 convergence rows and 340 raw edge-score rows; tests cover canonical-oracle identity, annotation groups, ranking ties, validation leakage, deterministic policy fitting, and budget/cost accounting.

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